# Grid GDP: GDPopt and MINLP reformulations

Target reached | Target not reached | Unsupported

| | GDPopt RIC | GDPopt GLOA | GDPopt LOA | Big-M BARON | Hull BARON |
|---|---|---|---|---|---|
| ackley | 899.10 | 909.69 | 310.47 | unsupported | unsupported |
| alpine1 | 0.88 | 0.85 | 0.95 * | unsupported | unsupported |
| alpine2 | 1789.14 T | 1789.40 T | 1795.11 T | unsupported | unsupported |
| bohachevsky2 | 175.17 | 223.44 | 154.58 | unsupported | unsupported |
| camel3 | 1800.10 T * | 1789.34 T * | 1789.64 T * | 1.15 | 1.00 |
| csendes | 0.92 | 0.94 | 1.21 | unsupported | unsupported |
| deflected corrugated spring | 507.80 | 440.76 | 378.11 | unsupported | unsupported |
| dropwave | 120.15 * | 176.69 * | 221.96 * | unsupported | unsupported |
| easom | 1176.78 | 1769.44 | 1787.95 | unsupported | unsupported |
| griewank | 1214.84 | 1581.92 | 1042.31 | unsupported | unsupported |
| holder table | 1148.15 | 1200.31 | 1356.64 | unsupported | unsupported |
| hosaki | 1569.06 | 1524.42 | 1758.69 | 1.30 | 1.29 |
| levy | 1731.14 | 1746.53 | 1794.67 T | unsupported | unsupported |
| levy13 | 1800.10 T | 1789.92 T | 1774.92 | unsupported | unsupported |
| michalewicz | 1237.94 | 1283.66 | 1192.68 | unsupported | unsupported |
| rastrigin | 78.26 | 114.81 | 180.11 | unsupported | unsupported |
| rosenbrock | 334.94 | 278.13 | 655.39 | 0.76 | 0.81 |
| shubert | 1069.70 | 1037.66 | 976.26 | unsupported | unsupported |
| styblinski tang | 1339.67 | 1440.10 | 1562.35 | 1.25 | 0.95 |
| sum of squares | 822.85 * | 754.81 * | 851.15 * | 0.55 | 0.58 |
| xin she yang3 | 1.01 | 0.90 | 1.37 | unsupported | unsupported |

Cell values: measured completion time (s). T: time limit; *: corrected Pyomo profile.

Colors show target attainment within 1e-5 of the CSV minimum, not an optimality certificate.